

ST2334 Cheat Sheet, AY26/27 S1

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1 Basic Concepts of Probability

1.1 Experiments, sample spaces, and events

- **Statistical Experiment** is a procedure that produces an observation.
 - **Sample Space** is the set of all possible outcomes of a statistical experiment.
 - **Event** is a subset of the sample space.
 - **Sample Point** is a single outcome of a statistical experiment.
- Sample Point $\subseteq$ Event $\subseteq$ Sample Space

Sure Event and Null Event

- **Sure Event** = Sample Space, S
- **Null Event** = Empty Set / Complement of Sure Event

1.2 Event Operations & Relationships

- **Union** of A and B, $(A \cup B)$
 $A \cup B = \{x \in U \mid x \in A \vee x \in B\}$
- **Intersection** of A and B, $(A \cap B)$
 $A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$
- **Difference** of B minus A, $(B - A)$
 $B - A = \{x \in U \mid x \in B \wedge x \notin A\} = B \cap \bar{A}$
- **Complement** of A ($\bar{A}$)
 $\bar{A} = \{x \in S \mid x \notin A\}$
- **Mutually Exclusive or Disjoint** Events
 $A \cap B = \emptyset$

Contained and Equivalent

- **Contained C**
 $A \subset B \Leftrightarrow \forall x, x \in A \Rightarrow x \in B$
- **Equivalent =**
 $A = B \Leftrightarrow A \subset B \wedge B \subset A$

Subset Relations

No.	Law	Identities
1	Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
2	Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
3	Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4	Identity laws	$A \cap U = A$ $A \cap \emptyset = \emptyset$ $A \cup U = U$ $A \cup \emptyset = A$
5	Complement laws	$\overline{\overline{A}} = A$ $A \cap \bar{A} = \emptyset$ $\bar{\bar{U}} = \emptyset$
6	Idempotent laws	$A \cup A = A$ $A \cap A = A$

No.	Law	Identities
7	De Morgan's laws	$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$
8	Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

1.3 Counting Methods

Multiplication and Addition Principles

- **Multiplication Principle**
A sequence of r stages with $n_1, n_2, \dots, n_r$ choices has $n_1 n_2 \dots n_r$ outcomes.
 $n_1 \times n_2 \times \dots \times n_r = n_1 n_2 \dots n_r$
- **Addition Principle**
k non-overlapping procedures with $n_1, n_2, \dots, n_k$ outcomes have $n_1 + n_2 + \dots + n_k$ outcomes in total.
 $n_1 + n_2 + \dots + n_k = n_1 + n_2 + \dots + n_k$

Factorial, Permutations, and Combinations

- **Factorial** ($n!$)
 $n! = n(n-1)(n-2) \dots 2 \cdot 1$
- **Permutation** (P_r^n)
Selection and arrangement of r objects from n objects, where order matters.
 $P_r^n = \frac{n!}{(n-r)!}$
- **Combination** ($\binom{n}{r}$)
Selection of r objects from n objects, where order does not matter.
 $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

1.4 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative Frequency

Suppose an experiment is repeated n times and event A occurs m times. The relative frequency of A is given by:

$$P(A) = \lim_{\{n \rightarrow \infty\}} \frac{m}{n}$$

As n increases, the relative frequency of A approaches the probability of A.

Axiomatic Definition of Probability

A probability function P is a function that assigns a number P(A) to each event A in the sample space S, such that:

$$0 \leq P(A) \leq 1$$
$$P(S) = 1$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Properties of Probability

Prop.	Name	Statement
1.4	Empty set	$P(\emptyset) = 0$
1.5	Finite additivity	If $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(A_1 \cup \dots \cup A_n) = P(A_1) + \dots + P(A_n)$
1.6	Complement	$P(\bar{A}) = 1 - P(A)$
1.7	Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$
1.8	Inclusion-exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
1.9	Monotonicity	$A \subset B \Rightarrow P(A) \leq P(B)$

Prop.	Name	Statement
1.10	Union bound (Boole's inequality)	$P(A_1 \cup \dots \cup A_n) \leq P(A_1) + \dots + P(A_n)$

1.5 Conditional Probability

- **Conditional Probability** ($P(A \mid B)$)
Probability of event A occurring given that event B has occurred.

Definition of Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Simpson's Paradox

A phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. Or formally,

$$P(A \mid B \cap C_i) \geq P(A \mid B' \cap C_i) \forall i \text{ but } P(A \mid B) < P(A \mid B')$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)P(B) \text{ if } P(B) > 0$$
$$\text{or } P(A \cap B) = P(B \mid A)P(A) \text{ if } P(A) > 0$$

Inverse Probability Formula

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

1.6 Independence

Events A and B are independent, $A \perp B \Leftrightarrow$

$$P(A \cap B) = P(A)P(B)$$

Tautologies:

- Suppose $P(A) > 0$ and $P(B) > 0$. If $A \perp B$, then A and B are not mutually exclusive (aka $A \cap B \neq \emptyset$)
- Suppose $P(A) > 0$ and $P(B) > 0$. If A and B are mutually exclusive, then $A \not\perp B$.
- $S \perp A$ for all events A
- $\emptyset \perp A$ for all events A
- If $A \perp B$, then $A \perp B'$, $A' \perp B$, $A' \perp B'$
- Independence cannot be expressed in terms of venn diagram

1.7 Law of Total Probability

Suppose $A_1, A_2, \dots, A_n$ is a partition of the sample space S

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B \mid A_i)P(A_i)$$
$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

1.8 Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{i=1}^n P(B \mid A_i)P(A_i)}$$

2 Random Variable

Random Variable is a function that transform a sample from sample space to a real number.

$$X: S \rightarrow \mathbb{R}: s \in S \text{ and } X(s) = x \in \mathbb{R}$$

Range space of X is set of real numbers that satisfies

$$R_X = \{x \mid X(s) = x \forall s \in S\} : R_X \in \mathbb{R}$$

Subsets of Sample Space

Set of all sample such that the result of its random variable is x.

$$\{X = x\} = \{s \in S : X(s) = x\}, \{X = x\} \in S$$

Set of all sample such that the result of its random variable is in A.

$$\{X \in A\} = \{s \in S : X(s) \in A\}, \{X \in A\} \in S$$

2.1 Probability Distribution

Probability Mass Function (pmf)

Discrete random variable, the number of $R_X = \{x_1, x_2, \dots\}$ is finite or countable.

pmf, f(x) is defined as probability for $\{X = x\}$

$$f(x) = \begin{cases} P(X = x) & \forall x \in R_X \\ 0 & \forall x \notin R_X \end{cases}$$

Well-defined pmf for discrete random variable X:

1. $\forall x_i \in R_X, f(x_i) \geq 0$
2. $\forall x_i \notin R_X, f(x_i) = 0$
3. $\sum_{i=1}^{\infty} f(x_i) = \sum_{x_i \in R_X} f(x_i) = 1$

Let $B \in R_X$

$$P(X \in B) = \sum_{x_i \in B \cap R_X} f(x_i)$$

Probability Density Function (pdf)

Continuous random variable, $R_X \in [a, b]$ is an interval or a collection of intervals

Well-defined pmf for continuous random variable X

1. $\forall x \in R_X, f(x) \geq 0$
2. $\forall x \notin R_X, f(x) = 0$
3. $\int_{R_X} f(x)dx = \int_{-\infty}^{\infty} f(x)dx = 1$

For any a and b such that $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Consequently,

- $P(X = x_i) = \int_{x_i}^{x_i} f(x)dx = 0 \forall x_i$; Specific point is zero
- $P(a \leq X \leq b) = P(a < X < b)$: Endpoint doesn't matter

2.2 Cumulative Distribution Function

Cumulative Distribution Function (cdf), F(x) for any random variable X is defined as

$$F(x) = P(X \leq x) = \begin{cases} f(0) + f(1) + \dots + f(x) & \text{if X is discrete} \\ \int_{-\infty}^x f(t)dt & \text{if X is continuous} \end{cases}$$

Let $a < b, x \in R_x$

$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a -)$$
$$P(X = x) = f(x) = P(x \leq X \leq x) = F(x) - F(x -)$$

Let $F(x) = \int_{-\infty}^x f(t)dt$

$$f(x) = F'(x) = \frac{d}{dx} \int_{-\infty}^x f(t)dt$$

2.3 Expectation and Variance

Expectations

Expectations or mean of X is defined by

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i) = \int x f(x)dx$$

- μ_X may not be in R_X

Properties of Expectation

1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$ or $\int g(x)f(x)dx$

Variance

Variance of X is defined as σ_X^2

$$\sigma_X^2 = V(X) = E\left[(X - \mu_X)^2\right] = E(X^2) - [E(X)]^2$$
$$= \sum_{x \in R_X} (x - \mu_X)^2 f(x) \quad \text{or} \quad \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx$$

Standard deviation of X is defined as σ_X

$$\sigma_X = \sqrt{V(X)}$$

Properties of Variance

- $V(aX + b) = a^2 V(X)$

3 Joint Distribution

- **Two/n-dimensional random variable, (X, Y)** = Given $s \in S, (X(s), Y(s), \dots)$
- **Range Space $R_{X,Y}$** = $\{(x, y) \mid x = X(s), y = Y(s), \forall s \in S\}$
- **Discrete / Continuous** two-dimensional random variable depends if both random variable is countable and finite or not

Joint Probability Function

Joint probability (mass) function, jpf is defined by $(x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

Well-defined jpf X, Y of both discrete and continuous random variable satisfies:

1. $f_{X,Y}(x, y) \geq 0, \forall (x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0, \forall (x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
4. $P((X, Y) \in A) = \sum_{(x,y) \in A} f_{X,Y}(x, y) = \int \int_{(x,y) \in D} f_{X,Y}(x, y) dy dx$

- Let $g1(\cdot), g2(\cdot)$ be arbitrary functions, $g1(X), g2(Y)$ are independent
- $f_X(x) > 0 \Rightarrow f_{Y \mid X}(y \mid x) = f_Y(y)$

3.3 Expectation and Covariance

Expectation:

$$E(g(X, Y)) = \sum_x \sum_y g(x, y) f_{X,Y}(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dy dx$$

Covariance, $\text{cov}(X, Y)$

$$\begin{aligned} \text{cov}(X, Y) &= \sum_x \sum_y (x - \mu_x)(y - \mu_y) f_{X,Y}(x, y) \\ &= E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X \mu_Y \end{aligned}$$

Properties:

- $X \perp Y \Rightarrow \text{cov}(X, Y) = 0$, converse is not true
- $X \perp Y \Rightarrow E(XY) = E(X)E(Y)$
- $\text{cov}(aX + b, cY + d) = ac \cdot \text{cov}(X, Y)$
- $\text{cov}(X, Y) = \text{cov}(Y, X)$
- $V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{cov}(X, Y)$

Correlation Coefficient $\rho(X, Y)$

$$\begin{aligned} \rho(X, Y) &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \\ -1 &\leq \rho(X, Y) \leq 1 \end{aligned}$$

Properties: $V(X + Y) = V(X) + V(Y) + 2 \text{cov}(X, Y)$

- $V(X \pm Y) = V(X) + V(Y)$
- $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n) + 2 \sum_{j>i} \text{cov}(X_i, X_j)$
- $V(X_1 \pm X_2) = V(X_1) + V(X_2)$

3.1 Marginal and Conditional Distribution

Marginal Probability Distribution of X is defined as sum of $f(y)$ after fixing $X = x$

$$f_X(x) = \sum_y f_{X,Y}(x, y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

- $f_X(x)$ satisfies all properties of probability function

Conditional probability Function

Conditional Probability function of Y given $X = x$ is defined as

$$f_{Y \mid X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Properties:

- $f_{Y \mid X}(y \mid x)$ is defined only for x such that $f_X(x) > 0$
- $f_{Y \mid X}(y \mid x)$ is not a probability function and therefore, does not need to satisfy requirement of sum = 1

Applications:

- $P(Y \leq y \mid X = x) = \int_{-\infty}^y f_{Y \mid X}(t \mid x) dt$
- $E(Y \mid X = x) = \int_{-\infty}^{\infty} y f_{Y \mid X}(y \mid x) dy$ aka regression function

3.2 Independent Random Variables

Random variable X and Y are independent

$$X \perp Y \Leftrightarrow f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

$\Leftrightarrow$ both of the following holds

- $R_{X,Y}$ is a **Product Space** when probability function is positive $R_{X,Y} = \{(x, y) \mid x \in R_X, y \in R_Y\} = R_X \times R_Y$
- $\forall (x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = c \cdot g_1(x) \cdot g_2(y)$$

where g_1 depends only on x , g_2 depends only on y , and c is constant

Suppose X, Y are independent variable:

- $P(X \in A; Y \in B) = P(X \in A) P(Y \in B)$
- $P(X \leq x; Y \leq y) = P(X \leq x) P(Y \leq y)$